

ECN 140 Empirical Project Summary:

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April 18, 2026

Section 1: Using AI to Help Kickstart a Data Analysis Research Project

This section summarizes a conversation I had with Claude (Anthropic) in late February while watching the 2026 Winter Olympic Games. I hope to use this document to illustrate how an AI assistant can help you move from a broad research interest to a working research project.



Stage 1: Ask AI a Question

The conversation started with a single question: *“Is there any research about judge bias in ice skating competitions?”* This is triggered by the online discussions following Ice Dance Competitions in the Milan Winter Olympic where a French judge was accused of favoring a French pair over an American pair. It turned out nationalistic bias in figure skating judging is a well-studied problem in sports economics. But before chatting with Claude I had no knowledge about this established literature.

Following my question, Claude searched the web and synthesized findings across academic papers, sports analytics conferences, and recent news.

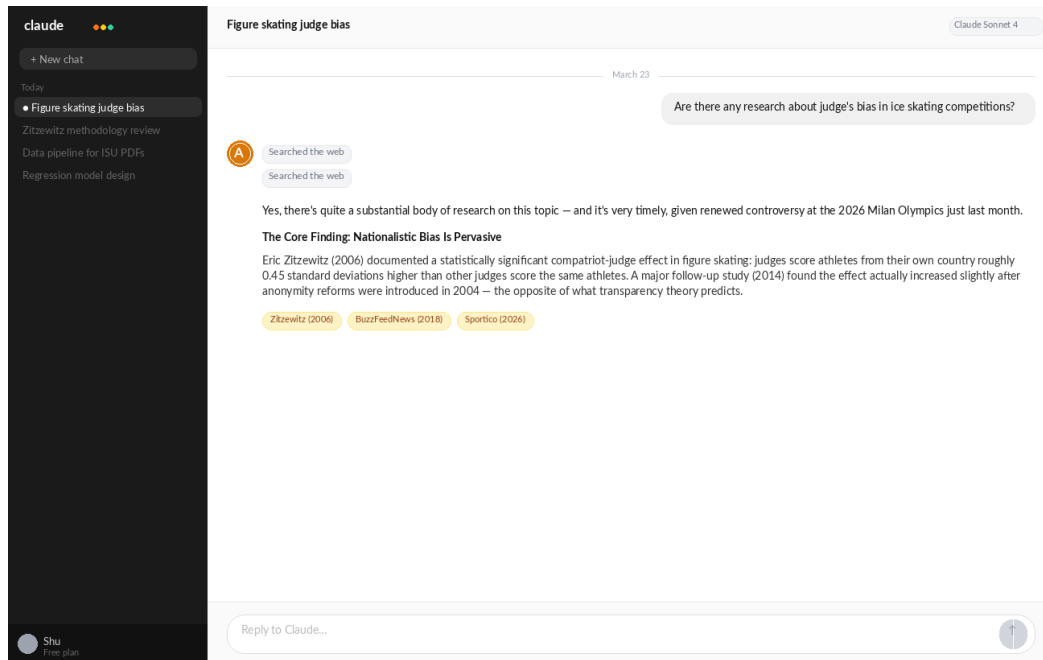


Figure 1. The opening exchange in Claude.ai.

The following sources were identified and summarized. I copied and pasted Claude's answer below.

Zitzewitz, E. (2006). Nationalism in Winter Sports Judging and Its Lessons for Organizational Decision Making. Journal of Economics & Management Strategy, 15(1), 67–99. — Foundational paper establishing the compatriot-judge effect. Same-country judges ranked competitors 0.45 standard deviations higher. Compared figure skating (national-federation-selected judges, biased) to ski jumping (centrally-selected judges, much fairer). Also found evidence of vote trading.

Zitzewitz, E. (2014). Does Transparency Reduce Favoritism and Corruption? Evidence From the Reform of Figure Skating Judging. Journal of Sports Economics, 15(1), 3–30. — Follow-up NBER working paper examining whether post-2002 ISU reforms worked. Counterintuitive finding: the compatriot-judge effect actually increased slightly after anonymity reforms, because shielding individual scores from outside scrutiny made corrupt behavior harder to detect.

Dumoulin, V., & Mercier, H. (2020). Accuracy and National Bias of Figure Skating Judges: The Good, the Bad and the Ugly. MIT Sloan Sports Analytics Conference. — Developed new mathematical tools for monitoring judge accuracy and bias. Applied to 2018 PyeongChang Olympics data. Found systemic bias: the two suspended Chinese judges were biased, but far from the worst offenders. Concluded the ISU's own monitoring was "utterly inadequate."

Sala, B., Scott, J., & Spriggs, J. (2007). The Cold War on Ice: Constructivism and the Politics of Figure Skating Judging. — Identified Cold War "bloc bias" in figure

skating from 1948 to 2002. NATO and Warsaw Pact nations judged along geopolitical lines; this effect diminished markedly after the Cold War ended.

BuzzFeed News (2018). The Edge: Figure Skating's Dirty Little Secret. Data and code available on GitHub. — Data journalism project applying Zitzewitz's methodology to 17 competitions (2016–2017) and the 2018 PyeongChang Olympics. Significant because the authors publicly released all data and code, making it reusable for subsequent research.

Claude also surfaced the 2002 Salt Lake City “Skategate” scandal, where French judge Marie-Reine Le Gougne admitted she had been pressured by her federation to favor the Russian pairs team in exchange for support for French ice dancers. This scandal directly motivated the ISU's 2004 judging overhaul, which is the institutional backdrop for all subsequent research.

Stage 2: Deep Dive on a Paper that Interests You

Prof. Zitzewitz's research seems to be the pioneering one in the literature. So I asked Claude to tell me a little more about it. I requested: *“Tell me more about Eric Zitzewitz's research on ice skating judge bias.”* Claude retrieved and synthesized his two main papers in detail. I copied and pasted Claude's answer below.

- *The compatriot-judge effect operates through two mechanisms: (1) direct favouritism — a judge scores their own skater higher — and (2) vote trading — other judges on the panel also score that skater higher when a compatriot judge is present, suggesting reciprocal deal-making.*
- *Bias increases with the stakes of the event. Judges do not simply show a constant mild preference; they amplify it specifically at the Olympics.*
- *The institutional mechanism is clear: national federations select their own judges, creating career incentives for those judges to favor their compatriots. In ski jumping, where the international federation selects judges centrally, the least biased judges are chosen.*
- *Countries ranked as more corrupt on general governance indices (e.g. Transparency International) also show stronger judging bias in figure skating.*
- *The 2004 anonymity reform made bias worse, not better — the opposite of what transparency theory predicts.*

This sounds very interesting. So I downloaded and read through the two papers. The empirical approach used is a (judge) fixed-effect model. Since we are going to talk about panel data and the fixed-effect model, I decided that this would be a promising direction for the ECN 140 empirical project.

Stage 3: Identifying a Research Gap

Since Zitzewitz's research used ice skating competitions that happened almost two decades ago, I asked Claude, *“Can you collect more recent years of data to repeat the analysis?”*

Claude identified the core methodological obstacle (individual judge scores were anonymised from 2004 to approximately 2018, preventing direct attribution), then mapped the available data sources, identifying the ISU official PDFs as the primary source for post-2018 coverage.

Stage 4: Verifying the Institutional Landscape

Before committing to a data collection plan, I step back and asks the Claude to first research in the institutional background. Although I watch ice-skating competitions from time to time, I do not follow the sport very closely. So it would be helpful to learn, before diving into data analysis, what has changed in the sport or institution since Zitzewitz’s original research was written. I asked Claude: “What major changes has the ISU made to its judging and scoring rules since Zitzewitz’s 2006 paper? Are there any changes that could affect a study of judge bias using post-2018 data?” Claude identified several changes. I copied and pasted Claude’s answer below.

- *The 2004 reform replaced the 6.0 scoring system with the International Judging System (IJS), which separates element scores (GOE) from component scores (PCS). Raw scores from before and after 2004 are not directly comparable.*
- *In 2018, the ISU expanded the Grade of Execution (GOE) scale from $-3/+3$ to $-5/+5$. Pooling GOE scores from both eras without standardisation would introduce a systematic scaling error.*
- *Individual judge scores were anonymised under the post-2004 rules, then partially de-anonymised starting around 2016 when the ISU began publishing detailed score sheets with judge positions. This shapes which seasons can actually be used: only post-2016 data includes the judge-level attribution needed for the Zitzewitz methodology.*
- *The COVID-19 pandemic cancelled the 2020 World Championships and severely disrupted the 2020–21 season. These gaps reduce the number of usable observations and should be noted as a limitation in any write-up.*

This conversation narrowed the usable dataset from “all post-2004 data” to “post-2016 data using the expanded GOE scale,” eliminating a major comparability problem.

Stage 5: From Plan to Working Code

After identifying the data sources, Claude produced a complete set of Python scripts that can directly download and compile a dataset for me to study judge bias in ice skating with more recent data. The following Table summarizes each script.

Table 1. Python Scripts for Data Collection, Compilation, and Analysis

Script	Purpose
<code>1_scrape_isu_pdfs.py</code>	Crawls the ISU website to discover and download official score sheet PDFs
<code>2_parse_isu_pdfs.py</code>	Extracts judge names, nationalities, and scores from PDFs using regex and pdfplumber
<code>3_build_dataset.py</code>	Merges all sources into a unified dataset and computes the core Zitzewitz measure (points vs. panel average)
<code>4_analyze_bias.py</code>	Runs all statistical tests: OLS with performance fixed effects, country rankings, season trend, discipline breakdown, and vote trading test

Stage 6: Setting Up Claude Cowork

The first five stages of this project all took place inside the Claude.ai chat interface in a web browser. Once the research plan was in place and the Python scripts were written, the next step

was to execute them. This is where Claude Cowork becomes useful. Cowork is a paid feature. It is not necessary but can help you save times in big data collection like our project, as is described below.

Stage 6A: Claude Pro Subscribers

Claude Cowork is available to Claude Pro subscribers (\$20 per month). Notice in Stage 5 that Claude wrote data scraping scripts in Python. With Claude cowork, you do not need Python installed on your computer. You can just open a Cowork session, point Claude at your project folder, and ask it to run the scripts. Claude will handle code execution, read error messages, and revise the code as needed. Claude Cowork also supports R, the language used in ECN 140. If you prefer to work in R rather than Python, you can ask Claude to write the data scraping scripts in R from the outset, and it will execute them the same way.

Stage 6B: Free-Plan Alternative

If you do not have a Claude Pro subscription, you can still complete the pipeline using the free Claude.ai chat interface in a web browser. The approach is more manual: create a dedicated project folder on your computer, paste the scripts Claude generates into that folder, and run them yourself. When a script produces an error or unexpected output, copy the error message into the chatbox and ask Claude to fix the code. The free-plan path requires a bit more effort and familiarity with your programming environment but is definitely manageable.

Step 1 — Install the Claude Desktop App

Claude Cowork runs inside the Claude desktop application. You need to download the app on your computer first. Once installed, open the app and look for the Cowork button in the sidebar.

Step 2 — Create a Project Folder

Before starting a Cowork session, create a dedicated folder on your computer for the project. For the figure skating project, this was a folder called “ECN 140 project” on my laptop. Place any existing code files, data downloads, or notes there. A single-purpose folder matters: Claude will only see what is inside it, and keeping it tidy prevents confusion about which files belong to the project.

Step 3 — Grant Claude Access to the Folder

When you open a Cowork session, Claude will ask which folder it should work in. Click “Select Folder” and navigate to your project folder. This grants Claude read and write access to that directory **only** — it cannot access files elsewhere on your computer without a separate permission grant. This is the key difference from the chat interface: Claude can now open, read, edit, and create files, which means it can run your scripts and save outputs without you typing a single terminal command.

Security note: when selecting a folder, navigate to the specific project directory rather than a broad parent folder such as Documents, Downloads, or your home directory. The narrower the folder, the smaller the set of files Claude can read or modify. It is good practice to create a fresh, empty folder for each project before starting a Cowork session — this way, you have a complete record of every file Claude creates or changes, and there is no risk of accidentally exposing unrelated work.

Step 4 — Run the Pipeline

With folder access granted, instruct Claude in plain language: “Run all scripts in the project folder in order.” Claude will inspect the folder contents, read the README, check for missing Python packages, and execute each script. If a script fails, Claude diagnoses the error and proposes a fix.

Stage 7: Auditing Data Quality

Once I had a compiled dataset, it was tempting to move straight to analysis. But data assembled from multiple sources — especially through web scraping and PDF parsing — almost always has errors, gaps, or inconsistencies that will distort results if left unchecked. So data quality check is very important. You can start with asking to do a systematic quality audit for you and then go to more specific requests like producing histograms of each variable, check duplicate rows of data and missing data.

For our project, producing histograms flagged a serious problem. The histogram of `judge_country` had a conspicuous spike that did not correspond to any real federation: tens of thousands of rows carried the literal string “ISU” in the `judge_country` column. This meant that a large block of the dataset was effectively missing the important judge-nationality attribute!

So I asked Claude to show which events, disciplines, and years the “ISU” rows were concentrated in and to cross-reference that with ISU rules on judge identity. It turned out that every World, European, Four Continents, and Olympic event from roughly 2017 onward was affected, while Grand Prix events were not. Claude then found the relevant ISU Communication, which mandates that judges’ names and federations be suppressed on the public judges-details protocols at championship events. So I had Claude null out every “ISU” value so that the missingness was explicit in the dataset, and I added a `judge_country_source` column so that any later backfill would be traceable to its origin.

Having such a large proportion of missing data in `judge_country` is not ideal for our data analysis. I asked Claude to search for more on ISU but nothing is found. This is strange, because during the Milan Olympics we do know the name of the French judge, and I’ve seen country-specific judge score break downs on the internet last month. So I opened my social media account and carried out a quick keyword search. I found that country-specific judge score break down pictures were screenshots taken from a website called [Skatingscores.com](https://www.skatingscores.com). Then I asked Claude to do a quick search on this website. It turned out that [Skatingscores.com](https://www.skatingscores.com) publishes per-event panels that list each seat number alongside the judge’s name and a two-letter country code. I had Claude write a scraping script to collect the information and fill in the missing `judge_country` information. After that, I asked Claude to re-run the original histogram and to confirm that no “ISU” values remained. There are still some remaining missing values in `judge_country`. But those were localized to events not covered by the external source.

Because [skatingscores.com](https://www.skatingscores.com) is an external source the ISU does not endorse, I asked Claude to run a validation exercise on the seat-to-judge mapping before trusting it in the final dataset. The check cross-references every piece of ISU-authoritative information we do have against the scraped panels. Basically, the seat-to-person binding for 2018/19-onward championships cannot be derived from any ISU-authoritative source currently available to us; [skatingscores](https://www.skatingscores.com) is a genuine external dependency. But every constituent the ISU does publish, that is the list of officiating judges, their federations, the set of panels per event, the scoring patterns of compatriot vs non-compatriot seats, is consistent with the mapping [skatingscores](https://www.skatingscores.com) provides. Rows backfilled from [skatingscores.com](https://www.skatingscores.com) are noted by having `judge_country_source = 'skatingscores'` in the dataset.

Stage 8: Monitoring Long-Running Jobs with Claude Dispatch

Scripts 1 and 2 in this pipeline — discovering and downloading ISU score-sheet PDFs across multiple seasons, then parsing them — can take hours. If you want to walk away from your computer you can use Claude Dispatch, a feature of the Claude mobile app that lets you hand off a long-running task, step away, and receive a push notification on your phone when the job finishes or encounters a problem that requires a decision.

Section 2: Empirical Project Instruction

The eight stages documented above show us how an initial question prompted by an Olympic broadcast can be developed to a research project with the help of AI. The final dataset is `judge_scores.csv`, which is uploaded to Canvas. The following table summarizes each variable provided in the dataset.

Table 2: A Comprehensive description of all 18 columns in the final dataset (863,130 rows, 8 seasons 2018/19 through 2025/26, 67 competitions, 753 skaters, 370 judges).

Variable	Type	Description	Values & Distribution
season	int64	Skating season, encoded YYYY where the two halves are the start and end calendar years (e.g., 1819 = 2018/19). Seasons run from late summer through the following spring.	1819: 121,470 1920: 100,167 2021: 37,543 2122: 132,566 2223: 105,726 2324: 113,175 2425: 118,455 2526: 134,028 8 distinct values; 0 nulls.
competition	object (str)	Short code for the event. Prefix identifies the event series; year is the calendar year the event was held.	Prefixes: wc (Worlds), ec (Europeans), fc (Four Continents), owg (Olympic Winter Games), gpf (Grand Prix Final), wtt (World Team Trophy), gp<ccc> (Grand Prix host, e.g. gpjpn, gpcan, gpusa, gpfra, gpchn, gprus). 67 distinct values; 0 nulls. Top by rows: owg2022 (28,980), wc2021 (25,713), wc2019 (25,182), owg2026 (25,092), ec2020 (24,273).
discipline	object (str)	Skating discipline.	MEN: 266,424 WOMEN: 264,701 ICEDANCE: 175,564 PAIRS: 156,441 4 distinct values; 0 nulls.
is_junior	bool	Whether the segment is a Junior-level competition (Junior Grand Prix, Junior Worlds).	False: 843,825 (97.8%) True: 19,305 (2.2%) 0 nulls.

Variable	Type	Description	Values & Distribution
program	object (str)	Segment within the event.	FS (Free Skate — MEN, WOMEN): 383,158 SP (Short Program — MEN, WOMEN, PAIRS): 304,408 FD (Free Dance — ICEDANCE): 97,840 RD (Rhythm Dance — ICEDANCE): 77,724 4 distinct values; 0 nulls.
skater_name	object (str)	Skater or team identity as printed on the ISU protocol. For PAIRS and ICEDANCE the two partners' names are joined with ' / '.	753 distinct canonical skaters/teams; 0 nulls. Top by rows: Kaori SAKAMOTO (7,776), Charlene GUIGNARD / Marco FABBRI (6,910), Kevin AYMOZ (6,908), Deniss VASILJEVS (6,453), Matteo RIZZO (6,435), Riku MIURA / Ryuichi KIHARA (6,316).
skater_country	object (str)	Skater's representing federation as a three-letter ISU code.	63 distinct federations; 0 nulls. Top: USA 113,825 • JPN 99,230 • CAN 80,531 • FRA 55,883 • ITA 51,923 • RUS 49,631 • KOR 39,411 • CHN 33,615 • GER 24,255 • GEO 23,247.
judge_name	object (str)	Judge identity. Read directly from the PDF for non-anonymized events; reconstructed from Skatingscores.com panels for anonymized championships (see Appendix in README for provenance and validation).	370 distinct canonical judges; 0 nulls. Top by rows: Salome CHIGOGIDZE (14,471), Nadezhda PARETSKAIA (12,071), Yury KLIUSHNIKOV (11,934), Agita ABELE (11,693), Anna KANTOR (11,131), Jung Sue LEE (9,447).
judge_count ry	object (str)	Judge's federation as a three-letter ISU code. Sourced per judge_country_source.	51 distinct federations; 0 nulls. Top: USA 66,232 • JPN 62,283 • CAN 59,717 • FRA 51,682 • ITA 50,994 • CHN 34,323 • GER 32,997 • KOR 32,612 • RUS 28,772 • GBR 25,193 • CZE 24,693 • GEO 24,043.
aspect_type	object (str)	Whether the row is an element Grade-of-Execution (GOE) or a Program Component Score (PCS).	GOE: 594,824 (68.9%) PCS: 268,306 (31.1%) 2 distinct values; 0 nulls.
aspect_name	object (str)	Element code (for GOE rows) or component name (for PCS rows).	1,744 distinct values; 0 nulls. GOE examples: 3A (triple Axel), 4Lz+3T (quad Lutz + triple toe loop combo), StSq4 (step sequence level 4), CCoSp4 (change-foot combination spin level 4), ChSq1 (choreographic sequence), FCSp4 (flying camel spin). PCS components: Skating Skills (69,958), Composition (69,958), Presentation (40,742), Transitions (29,216), Performance (29,216), Interpretation of the Music (22,562), Interpretation of the Music/Timing (6,654, ice dance only).

Variable	Type	Description	Values & Distribution
score	float64	Raw judge mark on the row's aspect.	GOE range: integers in [-5, +5] (11 values). PCS range: multiples of 0.25 in [0.25, 10] (40 values). Overall: 42 distinct values; 0 nulls; mean 3.14, median 3.
judge_count ry_source	object (str)	Provenance of the judge_country value, so analyses with strict provenance requirements can filter.	isu_pdf (directly from the PDF): 401,291 (46.5%) isu_pdf_gp_registry (GP officials lookup): 249,787 (28.9%) isu_comm_2735 (current ISU registry): 132,118 (15.3%) isu_comm_2663 (prior ISU registry): 52,009 (6.0%) skatingscores (external, validated): 25,642 (3.0%) event_time_correction (re-keyed to the federation the judge represented on the event date, for judges whose ISU-registered federation changed mid-panel dataset): 2,283 (0.3%) 6 distinct values; 0 nulls.
event_type	object (str)	Whether the segment is an individual competition or a team event.	INDIVIDUAL: 842,961 (97.7%) TEAM: 20,169 (2.3%) 2 distinct values; 0 nulls.
judge_number	float64 (integer-valued)	Seat number on the judging panel within a segment. Identifies one of the 9 (or occasionally 7 or 8) judges consistently across all skaters in that segment.	Seats 1 through 7: 96,571 rows each. Seat 8: 94,415 rows. Seat 9: 92,718 rows. Min 1, Max 9; 9 distinct values; 0 nulls.
element_index	float64 (integer-valued)	Order of the element within the skater's program, 1 being the first scored element. Set to 0 as a sentinel on PCS rows (which are program-level, not element-level).	0 (PCS sentinel): 268,306 1: 67,962 • 2: 68,233 • 3: 68,829 • 4: 68,144 • 5: 68,624 6: 59,990 • 7: 59,952 • 8: 31,796 • 9: 31,661 10: 27,242 • 11: 24,257 • 12: 18,080 • 13: 54 14 distinct values (0-13); 0 nulls.
compatriot	float64 (0 or 1)	Indicator: 1 if judge_country == skater_country for this row, 0 otherwise.	0: 794,251 (92.0%) 1: 68,879 (8.0%) 0 nulls.
points_vs_avg	float64	The exclude-self bias measure used throughout the analysis: score minus the NaN-aware mean of the other judges' scores on the same (competition, discipline, program, skater_name, aspect_name) group.	Mean: 0.000 (by construction). Median: 0.000. Overall range: [-9.75, +10.00] across all aspect types. Typical range (GOE rows): roughly [-3, +3]; std = 0.78. 0 nulls; 567 distinct float values.

Feel free to use this dataset, or a subset of this dataset to write your essay for this empirical project. Use analysis skills we learnt in ECN 140 such as fixed-effects models, hypothesis testing, data visualization to write the essay.

Alternatively, you can pick a topic that interests you and ask AI to find data for you. Start with a question, ask Claude to map the literature, identify a gap or an extension worth testing, and let the conversation guide you toward a workable design. The most important step is the first one: ask a question you actually care about.